RV College of Engineering®

Design Thinking

Department of ISE

Phase I

Title of the Project

Digitally Decoding Heritage



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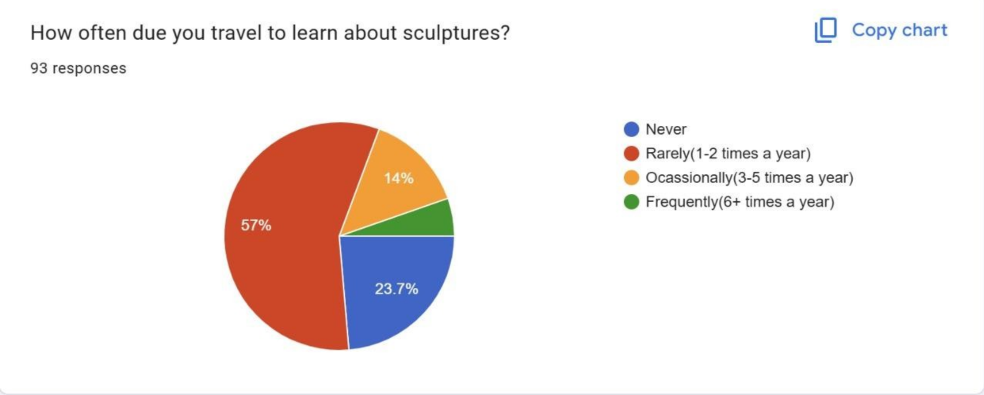
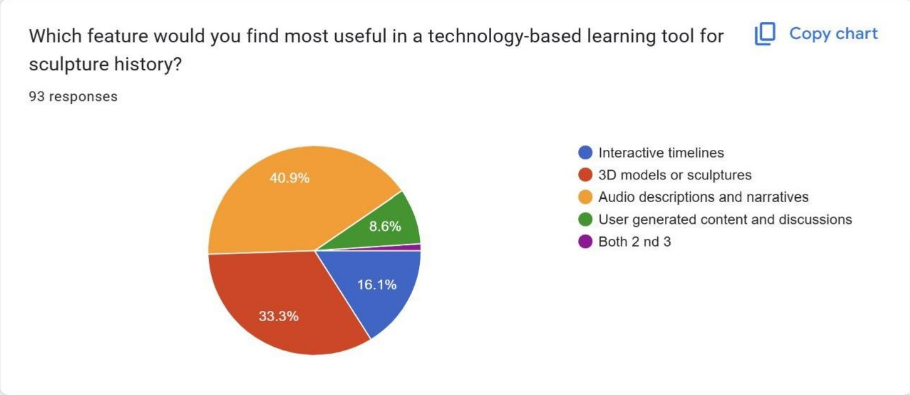
Motivation

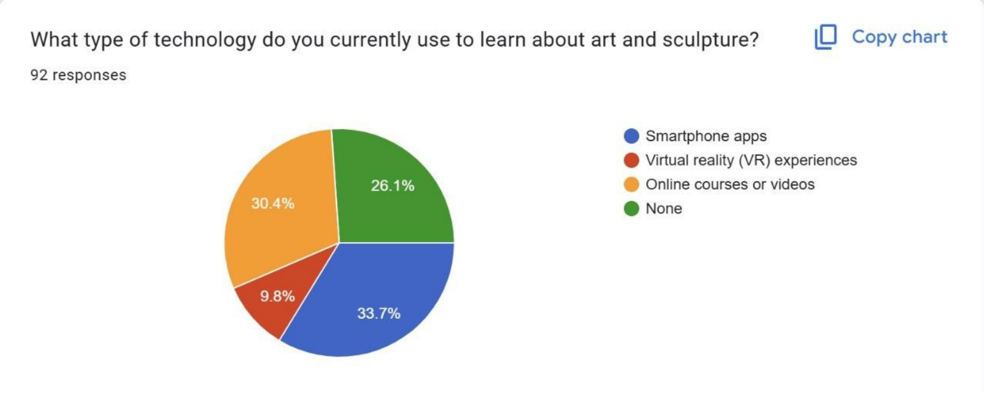
Go, change the world

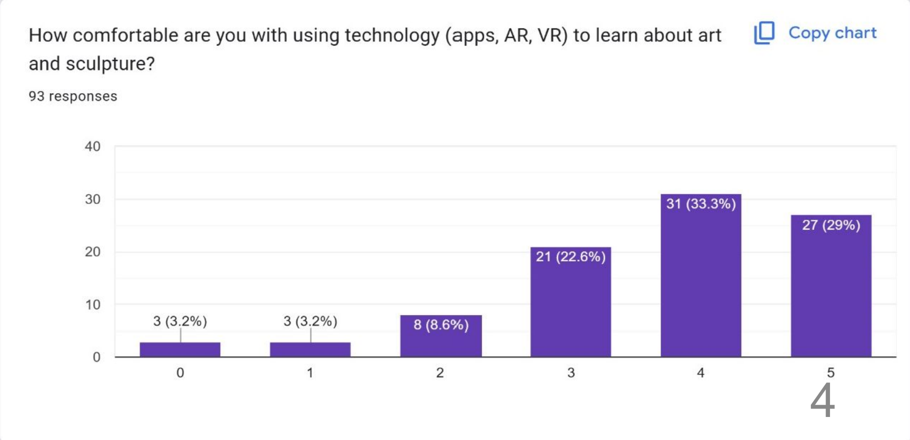
- **Replacing Traditional Guides** :Tourists often rely on guides who may provide inconsistent or incomplete information. This GUI offers a reliable, standardized source of knowledge.
- **Enhanced Accuracy** : By using verified data sources, the system ensures precise and comprehensive information about sculptures and their historical or cultural significance.
- **Engaging Learning Experience**: Interactive categorization and pop-ups make learning enjoyable, especially for younger audiences or those new to the temple's history.
- **Accessibility**: This system can cater to diverse audiences, including those with language barriers, by providing multilingual support and visual aids.
- **Preservation of Cultural Heritage**: Digitally archiving and categorizing sculptures helps preserve information about them for future generations.
- **Scalability**: The system can be extended to other temples or heritage sites, making it a versatile tool for cultural tourism.
- **Real-Time Exploration**: Users can quickly explore related sculptures within the same category without the need for manual navigation or external resources.
- **Self-Paced Tours**: Visitors can explore at their own pace, accessing information whenever they want, without being bound

QUESTIONAIRRE









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QUESTIONAIRRE

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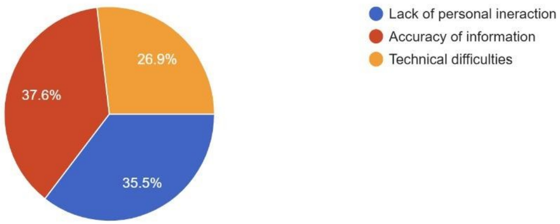
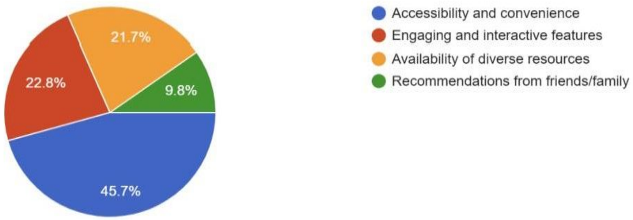
What concerns do you have about learning the history of sculptures through technology instead of traditional methods?

93 responses



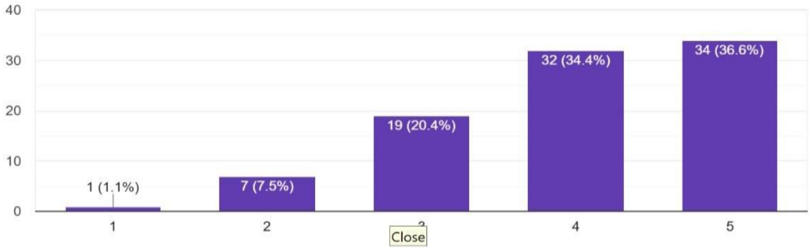
What motivates you to use technology for learning about the history of sculptures?

92 responses



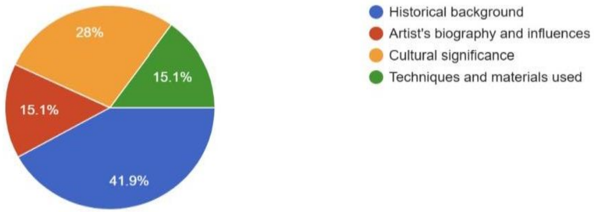
Would you prefer using an app that provides historical context and information about sculptures over reading traditional books or guides?

93 responses



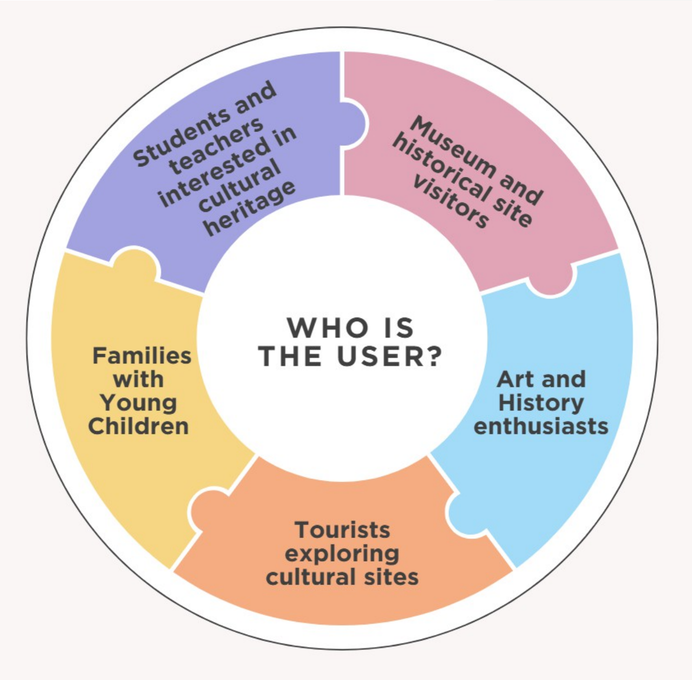
Which type of information would you prioritize when learning about a sculpture through digital platform?

93 responses



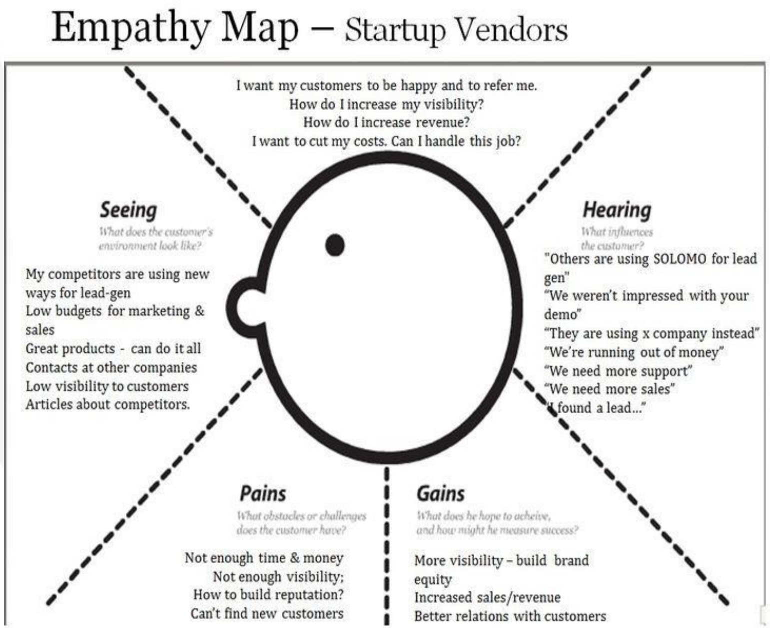
Empathy

Sample client details and Base/End user



Empathy Map

- List of addressable challenges
- List of pain points in addressing the need
- List of Gain points
- List of Insights



Literature Survey

- Highlight the articles/ publications referred.

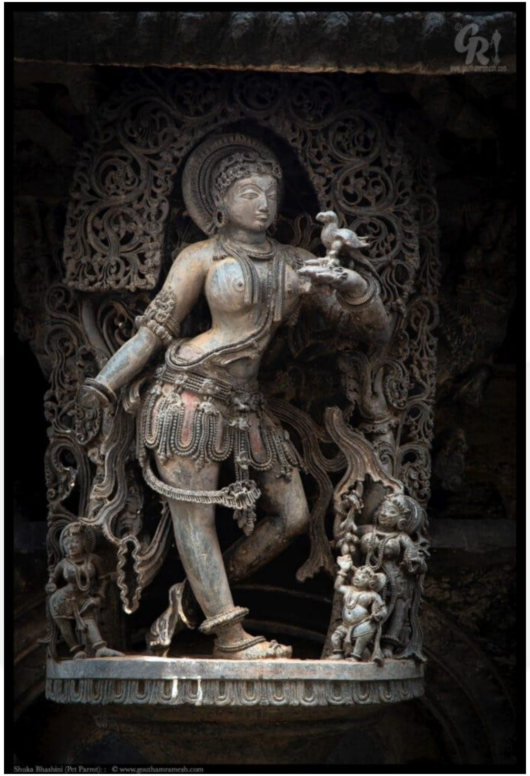
Problem Statement

The Beluru Chennakeshava Temple, renowned for its intricate sculptures and cultural heritage, often leaves visitors reliant on inconsistent guide information. This limits their understanding and access to related sculptures.

To address this, we are creating an interactive GUI system that allows users to:

- Scan sculptures using a camera.
- Access detailed, accurate information.
- View related sculptures through category-based pop-ups.

This solution modernizes exploration, offering an engaging and self- paced alternative to traditional guided tours while preserving the temple's heritage.



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Proposed Methodology

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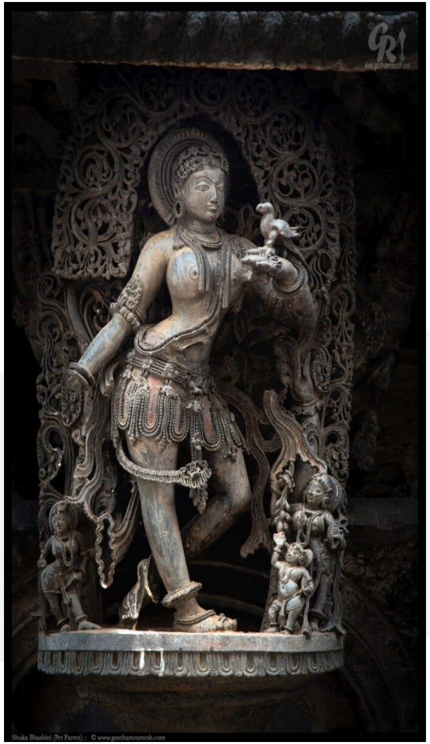
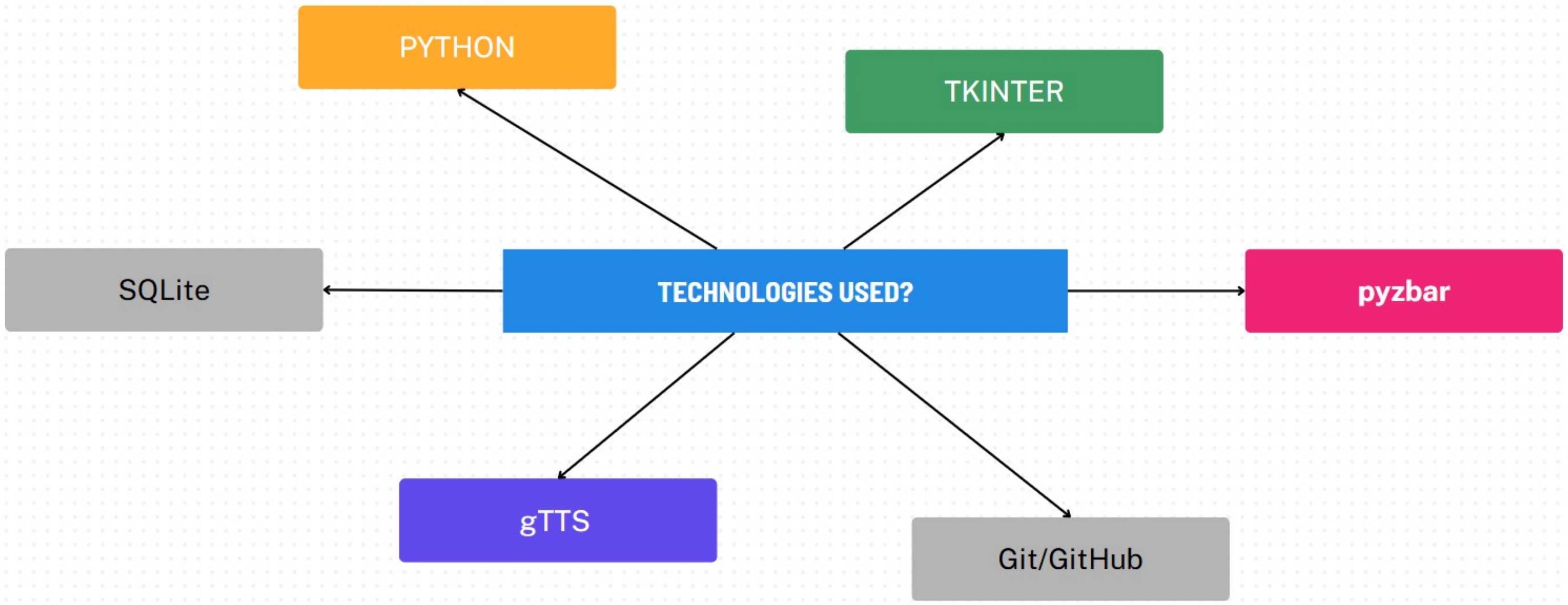
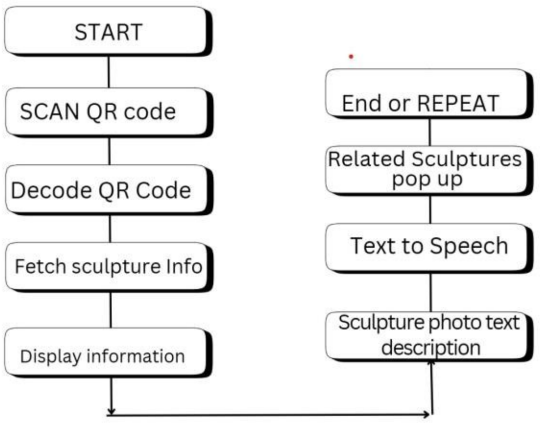


Prototyping Tools and Technologies

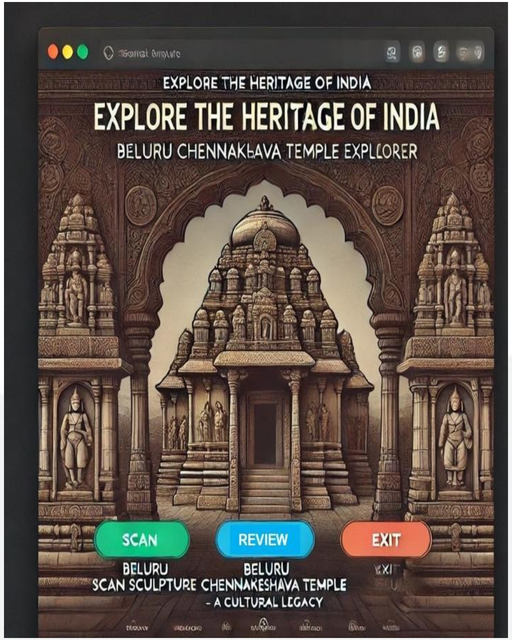


Expected Outcome

The system begins by scanning a QR code using a camera or scanner, decoding it with Pyzbar to retrieve a unique ID. This ID queries an SQLite database to fetch information about the sculpture, including its photo, description, and category.



The details are displayed using Tkinter, with an optional text-to-speech feature using gTTS for audio narration. A pop-up shows related sculptures from the same category, allowing users to explore further before repeating the process or exiting.



Home page of a GUI

References

- **Follow the IEEE Format for listing the References**